

# **RESEARCH METHODS FOR COMPUTER SCIENTISTS**

## **WRITING YOUR PROJECT REPORT**

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# Useful Video Tutorials and templates

How to format your work

- <https://blackboard.skopoku.org/tutorials/formatproj.mp4>

How to reference your work

- <https://blackboard.skopoku.org/tutorials/acawrite.mp4>

Project Proposal Template

- [https://blackboard.skopoku.org/tutorials/proposal\\_template.docx](https://blackboard.skopoku.org/tutorials/proposal_template.docx)

Project Report Title Page Template

- <https://blackboard.skopoku.org/tutorials/cover.docx>

# CHAPTER ONE - INTRODUCTION

# Writing the Chapter One - Introduction

- The aim of chapter one is to justify why the work you have done is important.
- You have to convince the reader that you have fixed a problem that existed.
- You also need to tell the reader what you aimed at in the project work and how the work was bounded.
- The reader should be motivated to read the other sections of the work after reading your chapter one
- The chapter focuses on the background of the study, the problem statement, the various objectives the study aims at achieving, the scope of the work and the significance of the project. It also gives a brief description of how the study has been organized

# Writing the Chapter One – Sections

- Provide an introduction that tells your readers what to expect in that chapter.

## 1.1 Background of the Study

## 1.2 Problem Statement

## 1.3 Objectives

***(Some supervisors require Aims and Objectives while others require General objective and Specific Objectives. Certain institutions research questions and/or objectives)***

## 1.4 Scope of the Study

## 1.5 Significance of the Study

## 1.6 Organization of the Work

# Writing the Chapter One – Background of the Study

- The background information provides sufficient content to the reader so they can understand the purpose of the work, the design and the interpretation of the results. It provides a general overview of the topic of study. It includes clear, focused, accurate and detailed information to understand the science investigated during the study
- **The Background generally consists of five elements:**
- **Element 1:** The writer first must state the broad theme or topic of the study.
- **Element 2:** Once the broad theme/topic has been introduced, its academic and practical importance (if applicable) must be explained. In short, you should provide a convincing answer to the question: “Why should anyone give a damn about this project?”.

# Writing the Chapter One – Background of the Study

- **Element 3:** The author next summarizes the available literature and cites the most important previous studies that are relevant to the current research. If an existing study were replicated, this should be clearly stated here. One should also include an in-text reference to the study that was replicated.
- **Element 4:** Next, the author indicates the most important gaps, inconsistencies and/or controversies in the literature that the current study will address. The author also explains the study's *main contribution* in such a way that the benefits to the reader are accentuated.
- **Element 5:** Finally, the context in which the study is being conducted must be stated.
- Some of the elements and sub-elements may be combined
- Study an example from:  
[https://blackboard.skopoku.org/tutorials/project\\_background.pdf](https://blackboard.skopoku.org/tutorials/project_background.pdf)

# Writing the Chapter One – Problem Statement

The statement of the problem is the focal point of your research. A concise **research problem** statement that, in one to three sentences, describes specifically *what the problem is that you intend to solve*. It explains *what problems or issues you wish to explore and why you wish to explore them*

It is just one sentence (with several paragraphs of elaboration).

It explains how the problem relates to business, social or political trends by presenting data that demonstrates the scope and depth of the problem.

Try to give dramatic and concrete illustrations of the problem. After writing this section, make sure you can easily identify the single sentence that is the problem statement



# Writing the Chapter One – Problem Statement

A well-written problem statement should capture the following elements:

**Element 1:** Provide a setting for your research.

**Element 2:** Provide research efforts (this is usually based on institutional requirement)  
(*For each research: identify purpose, methodology, results/finding and limitation*)

**Element 3:** State the gap(s) that exist that you want to solve.

**Element 4:** End by stating why the gap(s) have to be filled

- Study an example from:

[https://blackboard.skopoku.org/tutorials/project\\_problem.pdf](https://blackboard.skopoku.org/tutorials/project_problem.pdf)

# Writing the Chapter One – Problem Statement

**Some Types of Research Gaps** (The identified gaps provide a clear direction for future research):

## 1. Theoretical Research Gap

- **Example:** Examination of external factors influencing the adoption and effectiveness of AI in different organizational cultures.
- **Example:** Understanding the underlying reasons for the plateau in productivity gains due to AI over time.

## 2. Empirical Research Gap

- **Example:** Quantitative assessment of broader, organization-wide impacts of AI, beyond individual employee perspectives.
- **Example:** Analysis of causation between AI integration and organizational culture changes.

## 3. Methodological Research Gap

- **Example:** Deep analysis of specific AI technologies and their distinct contributions to operational efficiency.

## 4. Contextual Research Gap

**Example:** Comprehensive study on the impact of AI across various non-tech industries to understand broader implications.

## 5. Critical Research Gap

- **Example:** Investigating the impact of AI on diverse demographic groups within the workforce.

# Writing the Chapter One – Objectives

- The main aims that the project ought to achieve. Try to get between three and five numbered objectives. It should be specific (concise) and measurable
- Achieving your research objectives should lead you to solve the main research problem or answer the core research question or achieve the purpose of the study. Some of your objectives may be to test specific hypotheses, while others may relate to the gathering of descriptive data or to the development of a conceptual framework/model/system (or application).
- Your research objectives **should not focus on any managerial actions or activities** that may result from the study. It is, therefore, **incorrect to have research objectives in an academic study** such as the ones listed in the example below:
  1. To advise management on how best to improve the quality of service delivered to customers at Kumasi Technical University
  2. To develop an intervention plan in order to improve internal communication between top management and operational staff.

# Writing the Chapter One – Objectives Guidelines

- You should consider the following guidelines when drafting research objectives:
  - Research objectives must be presented in a bulleted list and should be phrased in the form: “To determine ...”, “To investigate ...”, “To evaluate ...”, “To compare ...”
  - Research objectives should be listed in order of importance or from the most general to the most specific.
  - Research objectives can be viewed as a set of “promises” made to the reader; promises that the researcher undertakes to achieve in the study. Be realistic in the promises you make! Limit the number of objectives you set by focusing your attention on three to five core objectives that will enable you to solve the research problem or answer the research question posed. Make sure that your study is properly designed to achieve these.
  - Each objective should focus on a single issue. Be careful for objectives containing the following words: “and”, “or”, “as well as”. These words often indicate a composite objective dealing with more than one issue.

# Writing the Chapter One – Objectives Guidelines

- Again, do not confuse research objectives with managerial objectives. Research objectives will focus on the things you want to do or achieve as a direct result of your research effort. As such, research objectives usually contain the following verbs in the format: *To <action-word><article> .....*
  - To analyse ...
  - To compare ...
  - To describe ...
  - To determine how many / which percentage ...
  - To develop a typology / classification scheme / system / conceptual framework ...
  - To identify ...
  - To investigate ...
  - To review the most recent literature ...
  - To test ...
  - To design
  - To examine
  - To assess

# Writing the Chapter One – Scope of the Study

The Scope includes the specific variables of study, the location of the study, the samples, the respondents and weaknesses of the study which are beyond the control of the researcher.

It is also the important section of a project which includes: coverage of the study, area, the subject, the research instruments, the research issues and concerns, the duration of the study and the constraints that have direct bearing on the result of the study.

Thus

- Write on what you ought to have done but you did not do and why.
- The scope in some institutions are termed as **Delimitation**

# Writing the Chapter One – Significance of the Study

- The benefit of the work you are doing to
  - the organization you are working on
  - the academic field
  - the society
    - The government in terms of policy
    - The consumers
- What are the benefit other students or developers derive from your work (theoretical and practical benefits). Outline any policies that can be obtained from your work

# Writing the Chapter One – Organisation of the Work

Provides brief description of the various chapters of the study.

## Example:

The first chapter focused on the background of the study, the problem statement, the various objectives the study aims at achieving, the scope of the work and the significance of the project. It also gives brief description of how the study has been organized.

The second chapter concerns itself with what other researchers have done concerning the topic under study. It reveals theories and concepts that have been generated. It focuses on the various technologies employed in the area under study.



# Writing the Chapter One – Organisation of the Work

The third chapter, on the other hand, focuses on the design of the proposed system. It designs how the components of the system interact to carry out the system functions. The methodology employed in this work is discussed in details.

The next chapter, mainly chapter four, implements the various algorithms generated in the third chapter. It ensures that all the objectives of the proposed system are achieved through implementation.

The last chapter, mainly chapter five ends the study with summary and discussion of the various findings. It also focuses on reviewing the strengths and limitations of the implemented system and finally provides direction into the future by stating research areas associated with the study

# **CHAPTER TWO**

# **LITERATURE REVIEW**

**Watch the Video Lecture:**

[https://blackboard.skopoku.org/tutorials/lit\\_review.mp4](https://blackboard.skopoku.org/tutorials/lit_review.mp4)

# Writing the Chapter Two – Literature Review

- A literature review is a means to an end, namely to provide background to and serve as motivation for the objectives and hypotheses that guide your own research.
- **A good literature review does not merely summarize relevant previous research.** In the literature review, the researcher critically evaluates, re-organizes and synthesizes the work of others. In a sense, compiling a literature review is like making a smoothie or fruit shake: The end product is a condensed mix that differs totally in appearance from the individual ingredients used as inputs
- The key to a successful literature review lies in your ability to “**digest**” information from different sources, critically evaluate it and present your conclusions in a concise, logical and “reader-friendly” manner.
- **Thus for each information you put down, ensure that the following question is answered at the end: “And so what?”**

# Concepts on Literature Review

- A literature review (i.e. review of relevant previous work) is expected to involve **accessing a selection of relevant previous work**, resources and materials with a strong relation to the research topic in question, **accompanied by a description AND a critical evaluation/critique and comparative analysis of each work**. It is **NOT a summary of available materials without any critical description**.
- The review of relevant previous work is expected to be comprehensive including a detailed documentation of significant prior research that should be comprehensive enough to demonstrate that you are aware of the major relevant sources of information.
- Most research projects arise out of considerable prior research, which should be summarized. ***You need to show the relationship between your research problem and this prior research and show where gaps exist in the knowledge area and by extension show how your work will fill this gap and by so doing make a unique/original or significant contribution to the body of knowledge/field of study in question***

# Concepts on Literature Review

- The literature review exercise should among other things:
  - review and document the results of other studies that are closely related to the proposed study
  - relate the proposed research to the larger, ongoing dialogue in the literature about a topic, filling in gaps and extending prior studies and work
  - provide a framework for establishing the importance of the study, as well as a benchmark for comparing the results of the proposed study with other previous findings.
  - demonstrate that you have a comprehensive grasp of the field and are aware of important recent substantive and methodological developments.
  - show how that the proposed study will refine, revise, or extend what is now known
  - provide a rationale for the proposed research work and demonstrate its lineage to existing knowledge, previous investigations, contemporary practice;
  - identify significant prior research questions and explain whether these questions have been asked before, and what answers have been obtained - i.e. outline, evaluate and synthesize current state of critical/ theoretical debate;
  - identify limitations of past/ current research, and explain your point of entry into the debate (identify gaps/ misinterpretation/ errors/ contradictions/ particular critical or theoretical problems)

# Concepts on Literature Review

**Try to answer the following questions when you read a material:**

- Do you agree with the arguments and conclusions of other researchers?
- If you disagree, why?
- Can you identify contradictory arguments or findings?
- How could one explain these contradictions?
- Do the findings of previous studies apply in all contexts or are the findings context- specific?
- What are the criticisms against the conceptual models or measurement approaches discussed in the literature?
- Which limitations should be considered when interpreting the results of previous research?

# Example: Definition of Impulse Buying

Assuming you are doing a project on impulse buying with the topic:  
**“Predicting the Impact of Impulse Buying on Academic Performance among Financially Unstable University Students”**

Your research will have to define impulse buying during the literature review.

**Question: How would you define impulse buying in your project**

**Note: A single statement of a definition of impulse buying is not enough as a literature review**

Study the acceptable definition from:

[https://blackboard.skopoku.org/tutorials/def\\_impulse\\_buying.pdf](https://blackboard.skopoku.org/tutorials/def_impulse_buying.pdf)

# Which Aspects Should I Include in a Literature Review?

- A good literature review should always include a discussion of the following aspects:
  - A brief discussion of where the specific topic under consideration fits into the “**bigger picture**” of the overall academic discipline (e.g., where does nudity appeals fit into the “bigger picture” of advertising appeals).
  - **Conceptual definitions** all the key concepts/constructs included in the study.
  - A focused and **synthesized discussion of relevant previous research findings** involving the constructs/concepts relevant to your study. Previous research may indicate:
    - Possible relationships between the chosen constructs (e.g., a correlation between communication satisfaction and job satisfaction),
    - Possible mediating (intervening) and/or moderating variables that influence the relationship between the chosen constructs,
    - Possible differences between groups on the chosen constructs (e.g., differences between males and females with regard to sensation seeking),
    - The contexts in which the constructs and relationships have previously been tested (e.g., among university students or in a specific industry),
    - Possible gaps, inconsistencies, controversies and/or unanswered questions in the literature that could form the basis of a new study,
    - The results of previous hypothesis tests involving the selected constructs/concepts or relationships, and
    - Possible untested hypotheses or propositions involving the chosen constructs.



# General Guidelines on Structuring a Literature Review

- A literature review is primarily structured through the use of headings, sub-headings and sub-subheadings.
- Consider the following principles when developing the headings of your literature review:
  - It is best to use a “funnel approach” when structuring a literature review. Start the review by placing the specific topic being discussed into an appropriate broader context and then focus your discussion on more specific issues.
  - Take time to carefully plan the headings and sub-headings that you will be using in order to ensure a logical flow of information. Mind-maps come in during this planning phase.
  - Headings should be descriptive and informative, that is, they should tell the reader exactly what is to be covered in the section to follow. Avoid headings that consist of a single word.
  - Make sure that the information you place under each heading are, in fact, related to and reflected in the heading.
- Study the example of literature review headings:

[https://blackboard.skopoku.org/tutorials/project\\_lit\\_heading.pdf](https://blackboard.skopoku.org/tutorials/project_lit_heading.pdf)

# **CHAPTER THREE METHODOLOGY OR PROPOSED SYSTEM DESIGN**

**Watch the Video Lecture:**

[https://blackboard.skopoku.org/tutorials/meth\\_end.mp4](https://blackboard.skopoku.org/tutorials/meth_end.mp4)

# Writing the Chapter Three – **Its Concept**

- A research methodology encompasses the way in which you intend to carry or carried out your research. You have to answer these three questions (WWH) in your chapter three:
  - **What** (methods did you use or you are going to use)
  - **Why** (did you choose these methods and leave the alternatives)
  - **How** (did you use these methods or you are going to use these methods)
- Your chapter three includes how you plan to tackle things like **system design, system implementation, data collection methods, statistical analysis, participant observations, and more.**
- If your project is about system development, then chapter three is entitled: **Proposed System Design**
- If you project is about research only or both research and system development, then chapter three is entitled: **Methodology**

# Structure of System Development Project (1/3)

## CHAPTER THREE PROPOSED SYSTEM DESIGN

### 3.0 Introduction:

Tell us what we should expect in this chapter

### 3.1 Software Design Methodology

**What** software design methodology used: prototype, waterfall, etc

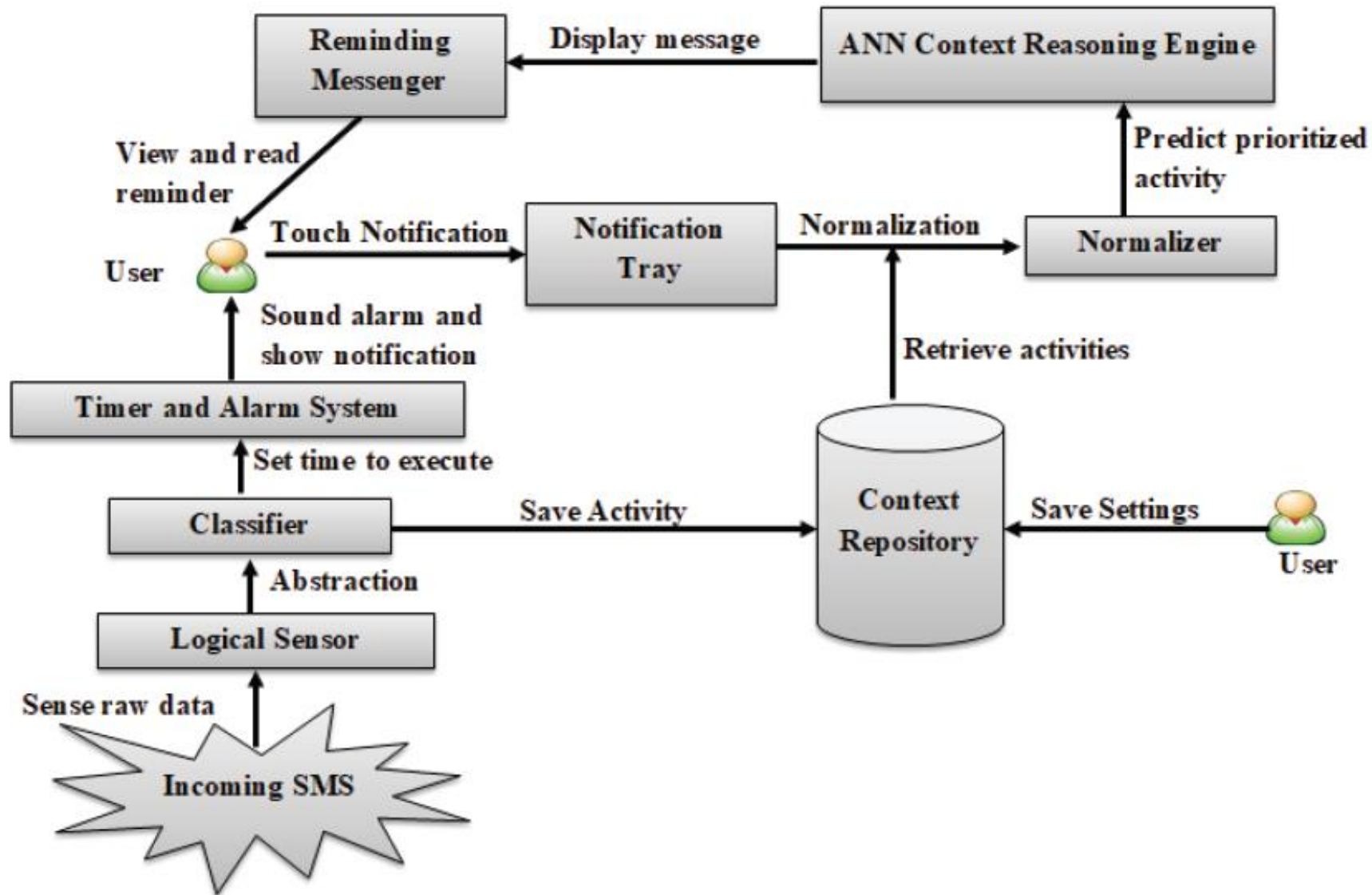
**Why** that software design methodology: Compared with other software designs and explain why you chose such design over the others. You may have to cite reference to support your argument

**How** did you use the steps in the software design methodology: For each stage in the chosen design method explain what you did in relation to your research work

### 3.2 System Architecture

Provide a model of the system and explain the what, why and how of the components

# Example of a System Model (2/3)



# Structure of System Development Project (3/3)

## 3.3 System Design

- Data flow diagram (DFD) and flowcharts /pseudocodes to describe the logic behind its operations
- Describe the various tables used, why you used them and represent them using Entity-Relation Diagram

## 3.4 User-System Interaction

Describe how the users interact with the system. Use diagrams like use case diagrams to illustrate

NB:

- You can create your subsections for each major section. It is advisable that you do not go beyond level three (Example of level three 3.5.1, 3.4.2 etc) You **should not** have **3.4.2.1 (ie level four)**
- Figures should be labeled as Fig 3.1, Fig 3.2, 3.3, 3.4 etc
- Tables should be labeled as Table 3.1, Table 3.2, 3.3, 3.4 etc

# Structure of Other types of Project (1/3)

## CHAPTER THREE

### METHODOLOGY

#### 3.0 Introduction:

Tell us what we should expect in this chapter

#### 3.1 Research Design and Instruments

**What** design and instruments are you using (W)

**Why** those design and instruments (W)

**How** did you use the design and instruments(H)

#### 3.2 Data Collection and Data Preprocessing

Describe your research population using WWH

Describe your Sampling technique or the attributes of the data for modeling using WWH

State data size and explain why it is enough for your study

Explain the ethical issues you followed during data collection

Explain how you handle (multi-)collinearity, missing data and/or unbalanced data

# Structure of Other Types of Project (2/3)

## 3.3 Model Development / Method of Analysis

Describe the following:

**What** methods (or machine learning algorithms) did you choose for the model (or data analysis)

**Why** did you choose those methods (support your reason with references from the literature)

**How** did you use those methods.

(**Note:** The How section of a model development should include a diagram of its architecture and how accuracy of the model is evaluated)

**NB:** If your project work also has software development in addition, then continue with 3.4 to 3.6

## 3.4 System Architecture

- As described above

## 3.5 System Design

- As described above

## 3.6 User-System Interaction

- As described above



# Structure of other types of Project (3/3)

NB 1:

- You can create your subsections for each major section. It is advisable that you do not go beyond level three (Example of level three 3.5.1, 3.4.2 etc) You **should not** have **3.4.2.1 (ie level four)**
- Figures should be labeled as Fig 3.1, Fig 3.2, 3.3, 3.4 etc
- Tables should be labeled as Table 3.1, Table 3.2, 3.3, 3.4 etc

NB 2:

- The subsections of the methodology section of a other types of project typically include:
  - **Participants, Materials, Design and Procedure**
- Ensure that the following areas are identified at the end of your chapter three:
  - **The type of research you did.**
  - **How you collected your data.**
  - **How you analyzed your data or developed the model**
  - **Any tools or materials you used in the research.**
  - **Your rationale for choosing these methods and how you used them.**

**CHAPTER FOUR**  
**RESULTS AND ANALYSIS**  
**OR**  
**SYSTEM IMPLEMENTATION AND TESTING**

# Structure of **System Implementation and Testing**

## CHAPTER FOUR

### SYSTEM IMPLEMENTATION AND TESTING

This chapter implements the various algorithms generated in the third chapter. It shows the various interfaces of the system. The various testing mechanisms employed and the appropriate error messages displayed with invalid entries and processes are also discussed in this chapter.

#### **4.1 Structure of Source Package**

Describe the various classes/packages of the application. Use class diagrams and/or other diagrams to describe the interaction of classes/files. Describe how you have organised your work into folders. State the files that are contained in each folder (You can add their functions). How the files in each folder interact. State the relationship between the folders (which folder rely on which ones)

#### **4.2 System Interfaces**

Describe the various interfaces of the systems (copy and paste the interfaces and write some descriptions – that is what the interface does and when to use it when the system deployed)

# Structure of System Implementation and Testing

## 4.3 Testing of the Implemented System

(Interfaces and write up – stating what happens when wrong entries are made)

### **Note:**

- Figures should be labeled as Fig 4.1, Fig 4.2, 4.3, 4.4 etc. Put the labels and captions at the bottom of the figure
- Tables should be labeled as Table 4.1, 4.2. 4.3. 4.4 etc. Put the labels and captions on top of the table

# Structure of **Results and Analysis**

- There is **no specific format** for Results and Analysis section. However, the following guidelines should be followed:
  - Brief introduction: This tells the reader what to expect
  - Description of the collected data (use descriptive statistics – tables, charts, averages (mean, mode, median) to describe the relevant feature of the data collected)
  - Present results of (multi-)collinearity, missing data and/or unbalanced data (where appropriate)
  - Present the results of the model developed (use charts and tables to describe the results during training and testing)
  - Analyse the results of the model and comment on its accuracy
  - If you project also involves software development the continue with **Structure of Source Package, System Interfaces and System Testing**

# Structure of Results and Analysis

## Note 1:

- The results should be presented in a clear, logical, and structured format, using tables, graphs, and charts where appropriate for clarity. This presentation makes it easier for readers to grasp the significance of the findings in relation to the research questions.
- Endeavour to demonstrate how the results address each specific research objective (or specific research question) outlined in the study.
- Give a factual account of the data collected and the model results, whilst the discussion should show how the results fill the gaps identified in the literature review and problem statement
- Do not speculate

## Note 2:

- Figures should be labeled as Fig 4.1, Fig 4.2, 4.3, 4.4 etc. Put the labels and captions at the bottom of the figure
- Tables should be labeled as Table 4.1, 4.2. 4.3. 4.4 etc. Put the labels and captions on top of the table

# **CHAPTER FIVE**

## **DISCUSSION, CONCLUSION AND FUTURE**

# Structure of Chapter Five (1/2)

## CHAPTER FIVE

### DISCUSSION, CONCLUSION AND FUTURE WORK

The final chapter ends the study with a discussion on the summary of the study. It also focuses on reviewing the implemented systems (or conceptual framework or model) and provides direction into the future by stating the research gaps or limitations associated with the study.

#### **5.1 Summary of the Study**

Discuss the summary of the study as follows: Briefly summarise the background and the problem statement in the first paragraph. For the subsequent paragraphs, pick each objective: state how you achieve it by describing briefly the method used, the results obtained and how it filled the gap identified (where appropriate)

#### **5.2 Analysis of the Implemented System (or Conceptual Framework or Model)**

Discuss your objective view of the artefact and state its strength and limitations



# Structure of Chapter Five (2/2)

## 5.3 Conclusion

Summarize the significance of the work.

Limitations of the study (Note: do not state the limitation of the artefact but the entire project such as time constraint, financial issues, hindrances to data collection. Discuss how that might affected the results).

*Optional: Provide recommendations for policy directions*

## 5.4 Future Work

From the limitations of the artefact and hindrances that affected the results of the study, state what should be done in the future to handle the various limitations. The aim is to fill a gap which has been created in your work. (*The gap to be addressed should be theoretical, empirical, methodological, contextual or critical in nature*)

# **ABSTRACT**

# Writing the Abstract of the Report

- The abstract is a short **summary** of a project work with a maximum length of **250 words**. Most readers first scan the abstract to decide whether reading the rest of the project work would be worthwhile. The abstract, therefore, serves as an important “window display” or “advertisement” for your work and provides an opportunity to impress the reader.
- The main problem with abstracts is that they are often so vaguely written that they do not grab the reader’s attention. One should always try to give the reader enough concrete information in an abstract to get them interested in your work

# Writing the Abstract of the Report

- An abstract should include the following seven elements:
- **Element 1:** The abstract must start with a brief theme sentence to orientate the reader about the overall issue addressed in the project. This sentence should grab the reader's attention.
- **Element 2:** The abstract should then indicate the main aim or purpose of the study.
- **Element 3:** Next, the academic and/or practical importance of the study should be explained.
- **Element 4:** The methodology used in the study should also be briefly described.
- **Element 5:** The main findings/results of the study should be summarized.
- **Element 6:** A statement of conclusions should indicate the contribution made by the study in filling gaps in the literature.
- **Element 7:** Finally, the practical or managerial implications of the study's findings should be highlighted **where appropriate**.

# Writing the Abstract of the Report – Example 1

**[Element 1]** Advertisements have become more suggestive as companies contest for consumer attention in an over-saturated market. One such suggestive approach is the use of celebrity appeals. **[Element 2]** This study investigates the influence of celebrity appeals on consumer attitudes towards the advertisement and the brand, as well as on consumers' intention to purchase the product. **[Element 3]** The results should assist marketers to ascertain whether celebrity appeals are effective or whether such appeals offend consumers. **[Element 4]** A survey of two thousand respondents aged 18 to 30 **[Element 5]** found that there is a significant correlation between gender and acceptance of celebrity content in print advertisements. In addition, advertisements containing celebrity appeals attracted attention and interest and were not perceived as particularly offensive. Advertisements containing clear celebrity interactions are more effective in attracting attention and being memorable than those with “ordinary” people. **[Element 6]** Celebrity appeals may be an appropriate, though controversial strategy to get the attention of more consumers. **[Element 7]** Marketers should, however, carefully evaluate the nature of the target market when considering celebrity appeals in advertisements.

# Writing the Abstract of the Report – Example 2

**[Element 1]** One of the main causes of death around the globe is malaria. Researchers have sought to develop predicting models for malaria outbreaks based on metrological data, climate data and the breeding cycle of plasmodium, the causative agent of malaria. **[Element 2]** This study predicts the severity of malaria based on environmental and biological factors. **[Element 3]** This study should assist researchers to determine more factors that contribute to the outbreak of malaria in certain areas. **[Element 4]** A logistic regression model was developed in this study with 417 respondents from the Bosomtwe District of Ghana to predict the severity of malaria based on such factors as sickle cell disease, stagnant water, garbage dump, wet lawns, and the use of treated mosquito nets **[Element 5]** with 83.3% accuracy rate. It was deduced that although children in the District are highly prone to malaria infection, the severity is very low. **[Element 6]** Environmental and biological factors also influence malaria infection. **[Element 7]** The study recommends that not just having a good sample size alone is important during machine learning model development but also having a good sample representation of the various class labels is equally important

# Tense Structure of Project Work Report

- Ideally,
  - Chapter 1 (introduction) should be **past tense**
  - Chapter 2 (literature) can be **present or past** depending on how you quote the material
  - Chapter 3 (methodology / proposed system design) should be **past tense**,
  - Chapter 4 (results and discussion / system implementation and testing) can be a combination of **past and present**
  - Chapter 5 (summary, conclusion and future work) should be **present and future**.

**Note: Always aim for consistency**

# Packaging Your Project Work Report

- TITLE PAGE
- DECLARATION
- DEDICATION (Optional)
- ACKNOWLEDGEMENTS
- ABSTRACT
- TABLE OF CONTENT
- LIST OF TABLES
- LIST OF FIGURES
- CHAPTER ONE – INTRODUCTION
- CHAPTER TWO – LITERATURE REVIEW
- CHAPTER THREE – PROPOSED SYSTEM DESIGN (METHODOLOGY)
- CHAPTER FOUR – SYSTEM IMPLEMENTATION AND TESTING (RESULTS AND ANALYSIS)
- CHAPTER FIVE – DISCUSSION, CONCLUSION AND FUTURE WORK
- REFERENCES
- APPENDICES



**THE END**